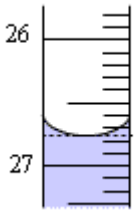
Useful data:

- Atomic weights:
  C=12, H=1, O=16, Na= 23, Cl=35.5, N=14, S=32, Fe = 55.8 , Co= 58.9, Cd=112.4
  Zn= 65.4, Hg= 200.6 g/mol
- R= 0.0821 atm.L/mol.K = 8.314 J/mol.K
- 1 atm = 760 mmHg.
- ___ K =_____ ^{o}C +273
- Density of vinegar solution = 1.03 g/mL
- Density of bleach solution = 1.08 g/mL
- Density of NaCl solution= 1.04 g/mL
- Specific heat of NaCl solution = 4.184 J/g.^{o}C



1. The safety notation that resembles an explosive dangerous is:
   a) I b) II c) III d) IV e) V



2) The best answer representing the volume measurement in the picture is:
   a) 26.22 b) 27.75 c) 27.70 d) 26.76 e) 26.5

3) What is the volume of a 12.2 g piece of metal with a density of 9.43 g/cm$_3$?
   a) 1.29 b) 2.35 c) 3.41 d) 2.47 e) 2.25

4) One of the following hydrated salts is not an alum:
   a) $NH_4Al(SO_4)_2.12H_2O$ b) $NaFe(SO_4)_2.12\ H_2O$ c) $Cu(SO_4).5H_2O$
   d) $NH_4Fe(SO_4)_2.12H_2O$ e) $KAl(SO_4)_2.12H_2O$

5) The data shown on the right were collected during water of crystallization experiment. If the molar mass of the anhydrous salt is 123 g/mol, then the value of X equals:

Mass of empty crucible	18.546g
Mass of crucible + hydrous salt	20.546g
Mass of crucible + anhydrous salt	19.611g

a) 6 b) 18 c) 5
d) 9 e) 7

6) **The following data were collected during molecular weight determination of a volatile liquid at 100 °C, the Mwt** (in g/mol) for this liquid equals:
a) 85.3 b) 81.6 c) 83.1
d) 88.2 e) 84.3

Mass of empty 200 mL flask + Al foil + rubber band	150.231 g
Mass of 200 mL flask + 30 mL volatile liquid + rubber band + Al. foil	165.213 g
Mass of flask + Rubber band + Al. foil + condensed vapor	150.813 g
Total volume of 200 mL flask	213.0 mL
Atmospheric pressure	754.6 mmHg

7) **During vinegar analysis, 25.0 mL sample of a diluted vinegas sample (dilution factor = 10) required exactly 18.20 mL of 0.20 M NaOH solution, the %by mass of acid in vinegar equals:**
a) 1.7% b) 85.0% c) 17.0% d) 12.7% e) 8.5%

8) **During bleach analysis, when the indicator was added, the solution became colorless, this means that::**
a) The endpoint of titration can be determined correctly and it is equal to the true value.
b) The endpoint of titration was determined correctly but it is not equal to the true vae.
c) You have passed the endpoint of titration, the % by mass of active material is equal to the true value.
d) You have passed the end point of titration, the % by mass of the active material will be lower than the true value.
e) You cannot know determine the endpoint of titration nor the % by mass of the active material. You should repeat the experiment.

9) **During the bleach analysis, the following data were collected, the % by mass of active ingredient in bleach sample equals:**
a) 43.2% b) 4.6% c) 6.4%
b) d) 7.7% e) 3.2%

Mass of KI	2.0 g
Initial burette reading	1.2 mL
Final burette reading	23.4 mL
Volume of 6M HCl solution	5.5 mL
Volume of original bleach	5.0 mL
Volume of indicator	3.2 mL

10) **Nitrogen and oxygen form an extensive series of oxides with the general formula N_xO_y. What is the empirical formula for an oxide that contains 63.64% nitrogen?**
a) N_2O b) NO c) NO_2 d) N_2O_3 e) N_2O_5

11) **The quantity of heat evolved from the reaction of 0.1 g of $Mg_{(s)}$ with an excess amount of 0.5 M $HCl_{(aq)}$ solution equals 2000 J. ΔH for the rxn $Mg_{(s)} + 2HCl_{(aq)} \rightarrow Mg^{2+} + 2Cl^- + H_{2(g)}$ per mol of $Mg_{(s)}$ equals:**
a) - 480 b) c) d) e)

12) **A sample of 1.7 g of NH_4NO_3 was mixed with 79.0 g of water (specific heat 4.184 $J.g^{-1}.°C^{-1}$) in a cup calorimeter (heat capacity 4.0 J/°C). If the initial temperature was 25.00°C and it became 23.09 °C upon dissolution. The enthalpy change of this process (in kJ/mol NH_4NO_3) equals:**
a) 30.07 b) 64.71 c) - 64.71 d) - 30.07 e) -7.20

13) One of the following equations represents the enthalpy of formation of $MgO_{(s)}$:

a) $Mg_{(l)} + ½\ O_{2(g)} \rightarrow MgO_{(s)}$ b) $Mg_{(s)} + ½\ O_{2(l)} \rightarrow MgO_{(s)}$

c) $Mg_{(s)} + ½\ O_{2(g)} \rightarrow MgO_{(s)}$ d) $Mg_{(s)} + ½\ O_{2(s)} \rightarrow MgO_{(s)}$

e) $Mg^{2+}_{(aq)} + H_2O_{2(l)} \rightarrow MgO_{(s)} + 2H^+_{(aq)}$

14) Concerning the kinetics of the reaction $3I^- + H_2O_2 + H^+ \rightarrow I_3^- + 2H_2O$, the rate of this reaction was equal to:

a) $\{\Delta[S_2O_3^{2-}]/\Delta t\}$ b) $2\ \{\Delta[S_2O_3^{2-}]/\Delta t\}$ c) $½\ \{\Delta[S_2O_3^{2-}]/\Delta t\}$ d) $2\{\Delta\ [I_3^-]/\Delta t\}$

e) $1/3\ \{\Delta[S_2O_3^{2-}]/\Delta t\}$

15) The rate constant (K) for a certain reaction equals $4 \times 10^{-4}\ s^{-1}$ at 200°C and $5 \times 10^{-2}\ s^{-1}$ at 400°C. The value of the E_a for this reaction is:

a) -63.9 KJ b) 639 KJ c) 6.39 KJ d) 63.9 KJ

16) When starch solution is added to a solution and the color of the solution turns to dark blue, this means that the solution is:

a) vinegar b) bleach c) $HCl_{(aq)}$ d) $NaOH_{(aq)}$ e) $I_{2(aq)}$ solution

17) One of the following is temperature dependent:

a) K b) E_a c) A d) R e) Mwt

18) One of the following equations represents enthalpy of dissociation of the weak acid:

a) $H_2O \rightarrow H^+ + OH^-$ b) $NaOH + HCl \rightarrow NaCl + H_2O$

c) $NaOH + CH_3COOH \rightarrow CH_3COO^-Na^+ + H_2O$ d) $H^+ + CH_3COO^- \rightarrow CH_3COOH$

e) $CH_3COOH + H_2O \rightarrow CH_3COO^- + H_3O^+$

19) The following data were collected for the reaction $H_2O_2 + I^- + H^+ \text{=====} \rightarrow I_2 + H_2O$ (unbalanced) at 25 °C:

trial	$[H_2O_2]$	[KI]	$[H^+]$	$[S_2O_3^{2-}]$	Time (*s*)
1	0.2	0.1	0.2	0.01	2.38
2	0.4	0.1	0.2	0.01	1.17
3	0.2	0.3	0.2	0.01	0.80
4	0.4	0.1	0.4	0.01	1.16

2.1×10^{-3}

The rate of this reaction for trial 1 equals:

a) 1.04×10^{-4} b) 7.12×10^{-5} c) 4.20×10^{-3} d) 3.50×10^{-5} e) 7.18×10^{-5}

20) The reaction is _ _ _ _ _ _ _ _ _ _ order overall.

a) zero b) second c) first d) third e) fourth

21) The following results were collected in order to determine the rate law for the reaction

$3A + 2B \rightarrow$ Products

The rate of this reaction is:

$[A]_\circ$	$[B]_\circ$	Initial rate of Product
0.030	0.02	4×10^{-5}
0.015	0.02	2×10^{-5}
0.015	0.04	4×10^{-5}

a) $R = k\ [A]^2\ [B]$ b) $R = k\ [A]^3\ [B]$

c) $R = k\ [A]\ [B]$ d) $R = k\ [A]\ [B]^3$

e) $R = k\ [A]\ [B]^2$

22) Which of the following statements is <u>correct</u>

a) The color of I_2-Starch complex is brown.
b) Rate of reaction does not depend on temperature.
c) Rate constant is temperature independent.
d) A blue color appears during the experiment when concentration of $S_2O_3^{2-}$ becomes zero.
e) Activation energy is the minimum energy required to increase the rate of reaction.

GOOD  LUCK

